MILITARY SPECIFICATION

ELECTRONIC COMPONENT ASSEMBLY: 11737807 (PRE-REGULATOR)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic assembly known as Electronic Component Assembly: 11737807 (Pre-regulator) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment; Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control System, Procedure for

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MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F11737807	Electronic Component Assembly: 11737807 (Pre-regulator)

PACKAGING DATA SHEET

11737807	Packaging of Electronic Component Assembly: 11737807 (Pre-regulator)

(Copies of specifications, standards drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The Electronic Component Assembly - (Pre-regulator), herein referred to as the assembly, shall be manufactured in accordance with Drawing F11737807 and drawings pertaining thereto.

3.2 General specification. The assembly shall meet the following requirements, where applicable, of MIL-F-13926:

a. Order of precedence

b. Dimensions and tolerances

c. Part identification and marking.

d. Electrical and electronic assemblies

e. Workmanship.

3.2.1 Soldering. Soldering shall be in accordance with MIL-S-45743 or MIL-S-46844, except that bare copper wire exposure shall be permitted at clipped ends of component leads.

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (^{o}F) and 90°F.

3.3.1 Loads, power, and auxiliary equipment. The assembly shall perform as specified herein when the loads, power, and auxiliary equipment of table I are applied as specified.

3.3.1.1 Full load output voltage characteristics. Under the full load (item 1.1) condition of table I, with the +29.0 Vdc power source (item 2.2 of table I) connected to pins as specified therein and set to apply +29.0 ±1.0 Vdc, and with the adjustable voltage source (item 2.1 of table I) set to apply +24.0 ±2.0 Vdc at J1-1, the full load output voltage characteristics at the designated terminals shall be as specified in table II.

3.3.1.2 Reduced load output voltage characteristics. Under the reduced load (item 1.2) condition of table I, with the +29.0 Vdc power source (item 2.2 of table I) connected to pins as specified therein and set to apply +29.0 ±1.0 Vdc, and with the adjustable voltage source (item 2.1 of table I) set to apply +24.0 ±2.0 Vdc at J1-1, the reduced load output voltage characteristics at the designated terminals shall be as specified in table II.

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TABLE I
LOADS, POWER, AND AUXILIARY EQUIPMENT

Item	Conditions	Characteristics	Connections
1.	Loads		Connected between the following pins, as specified:
1.1	Full loads		
1.1.1	Resistor	5Ω ±5 percent	J1-7 and J1-5
1.1.2	Resistor	27Ω ±5 percent	J1-9 and J1-5
1.2	Reduced loads		
1.2.1	Resistor	15Ω ±5 percent	J1-7 and J1-5
1.2.2	Resistor	39Ω ±5 percent	J1-9 and J1-5
2.	Power sources		Applied between the following pins, as specified:
2.1	+17 to +31 Vdc	Adjustable Ripple: 0.1 V rms maximum Source impedance: ≤ 0.01 ohm	J1-1(+) and J1-5(-)
2.2	+29.0 Vdc	Tolerance: ±1 Vdc Ripple: 0.1 V rms maximum Source impedance: ≤ 0.1 ohm	J1-3(+) and J1-5(-)
3.	Auxiliary equipment	Pre-regulator Switch Drive as shown on Figure 1, or equivalent.	Connected to J1 as shown on Figure 1.

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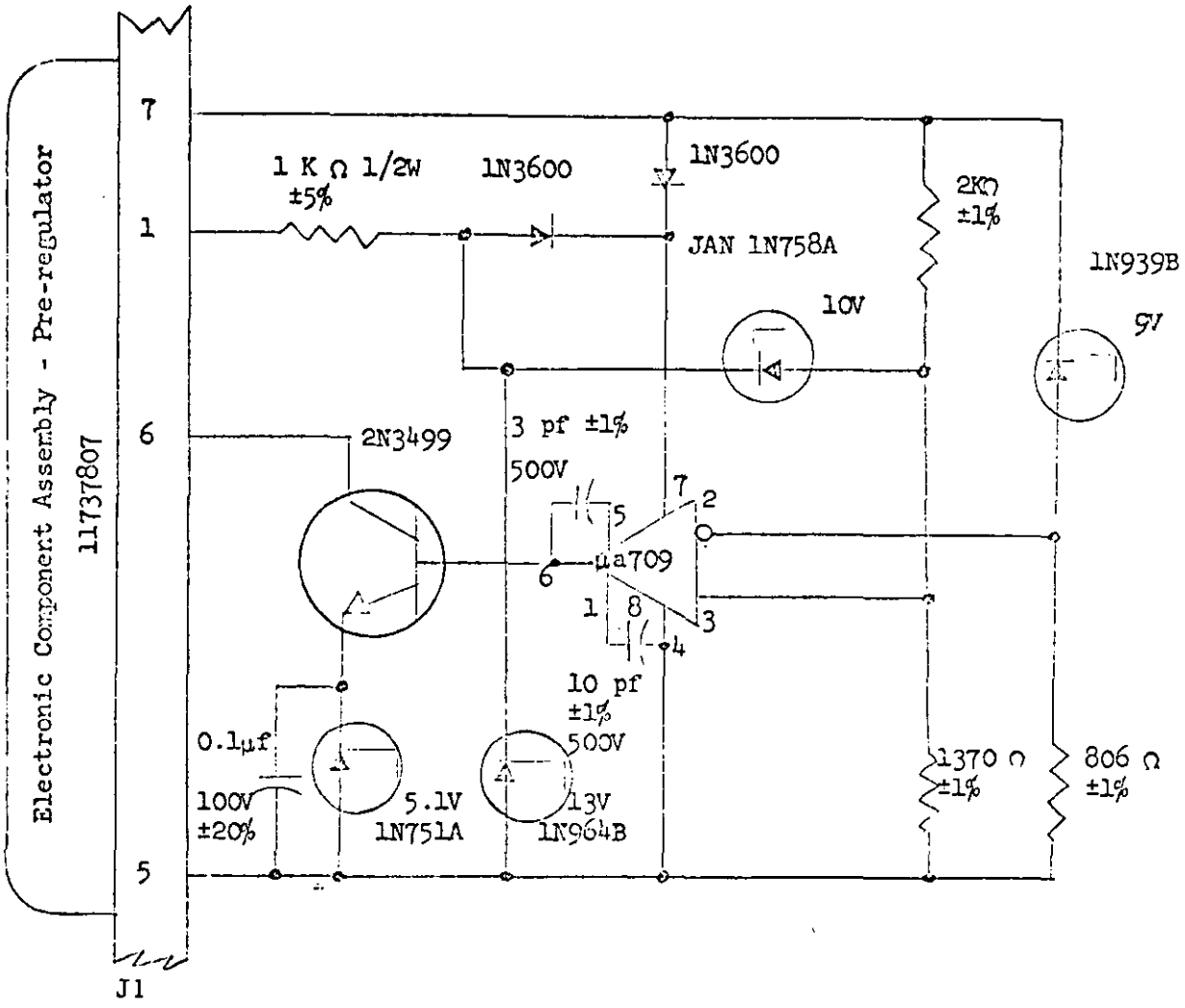


Figure 1. Pre-regulator Switch Drive Schematic

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TABLE II
OUTPUT VOLTAGE CHARACTERISTICS

Item	Pins of J1	Output voltage characteristics	
		DC voltage	Ripple
1	7	+15.0 ±1.2 Vdc	0.9V p-p
2	9	+10.0 ±0.8 Vdc	0.4V p-p

3.3.1.3 Full load output voltage regulation. Under the full load (item 1.1) condition of table I, and with the +29 Vdc power source (item 2.2 of table I) connected to pins as specified therein and set to apply +29 ±1 Vdc, the output voltage at J1-7 shall not change by more than plus or minus 0.5 Vdc when the voltage applied by the adjustable power source (item 2.1 of table I) to J1-1 is adjusted in turn to $+18.0\ ^{+0.6}_{-0}$ Vdc, +24.0 ±2.0 Vdc, and $+30.0\ ^{+0}_{-1.0}$ Vdc.

3.3.1.4 Continuity. Direct continuity (resistance equal to or less than 0.3 ohm) shall exist between J1-5 and each of J1-2, J1-4, and J1-8.

3.3.2 Environmental

3.3.2.1 Storage temperature. The assembly shall meet the requirements of 3.3.1 at ambient temperature after exposure to and thermal stabilization at -80°F and +160°F.

3.3.2.2 Operating temperatures. The assembly shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which the requirements of 3.3.1 shall be met at ambient temperature.

3.3.2.3 Vibration. The assembly shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

a. Amplitude: 1/64 inch, constant (1/32 inch double amplitude)

b. Vibratory motion: Simple harmonic

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c. Sweep time:	5 to 55 to 5 Hertz in a five-minute period
d. Vibration cycling:	Three linear or logarithmic sweeps applied to each of the three mutually perpendicular axes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three assemblies. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification. A record of this inspection, including certificates of conformance for materials, shall be submitted with the sample.

4.2.2 Contractor testing. When the contractor is responsible for conduction first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection and sampling procedures herein and submission of product provisions of MIL-STD-105.

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4.3.2 Examination and tests

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the respective specifications and Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Functional tests. The requirements and tests in table III shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature.

4.4 Special sampling. The selection of samples shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One assembly shall be selected at random by a Government representative as a control sample from each 50 produced. The control sample shall meet the requirements and tests in table IV.

4.4.2 Environmental. Three assemblies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in table V.

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table VI shall be utilized in performing the tests specified herein.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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TABLE III
FUNCTIONAL TESTS

Classification	Characteristics	Requirements	Test Paragraph
101.	Full load output voltage characteristics	3.3.1.1	4.6.2
102.	Reduced load output voltage characteristics	3.3.1.2	4.6.3
103.	Full load output voltage regulation	3 3.1.3	4.6.4
104.	Continuity	3.3.1.4	4.6.5

TABLE IV
SPECIAL SAMPLING - GENERAL

Classification	Control Tests	Requirements	Test Paragraph
301.	Fabrication	3.1	Applicable drawings-visual
302.	General specification	3.2	MIL-F-13926 - visual

TABLE V
SPECIAL SAMPLING - ENVIRONMENTAL

Classification	Control Tests	Requirements	Test Paragraph
303.	Storage temperature	3.3.2.1	4.6.6.1
304.	Operating temperatures	3.3.2.2	4.6.6.1
305.	Vibration	3.3.2.3	4.6.6.2

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TABLE VI
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1.	DC Power Supply	+17 to +31 Vdc, adjustable Ripple: 0.1 Vrms maximum Source impedance: ≤0.01 ohm
2.	DC Power Supply	+29 Vdc Ripple: 0.1 Vrms maximum Source impedance: ≤0.1 ohm
3.	DC Voltmeter	Range of 0 to 50 Vdc Accuracy: ±1 percent
4.	Oscilloscope	Frequency response: dc to 500 kHz Input impedance: 1 megohm or greater Sensitivity: 5 mV/cm Sweep time: 0.2 microseconds/cm
5.	Pre-regulator Switch Drive	As specified on figure 1, or equivalent
6.	Ohmmeter	Range: 0 to 10 megohms Accuracy: ±5 percent

4.5.2 Contractor furnished inspection equipment

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAPs forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4.6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of inspection equipment shall be so selected and controlled as to insure that the inspection equipment will reliability indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of inspection equipment.

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4.6 Test methods and procedures

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

4.6.1.1 Initial conditions. Connect the auxiliary equipment (item 3 of table I) to the assembly as specified therein.

4.6.1.2 Initial power conditions. Except as otherwise specified, connect the +29 Vdc power source (item 2.2 of table I) to pins as specified therein and adjust the source to apply +29.0 ±1.0 Vdc.

4.6.2 Full load output voltage characteristics test. Connect the full loads (item 1.1 of table I) to pins as specified therein. Connect the adjustable power source (item 2.1 of table I) to pins as specified therein, and adjust the source to apply +24.0 ±2.0 Vdc. Measure the output voltage characteristics at J1-7 and J1-9, to determine compliance with 3.3.1.1.

4.6.3 Reduced load output voltage characteristics test. Connect the reduced loads (item 1.2 of table I) to pins as specified therein. Connect the adjustable power source (item 2.1 of table I) to pins as specified therein, and adjust the source to apply +24.0 ±2.0 Vdc. Measure the output voltage characteristics at J1-7 and J1-9, to determine compliance with 3.3.1.2.

4.6.4 Full load output voltage regulation test. Connect the full loads (item 1.1 of table I) to pins as specified therein. Connect the adjustable power source (item 2.1 of table I) to pins as specified therein, and adjust the source to apply, in any order, the following voltages at J1-1: +18.0 $^{+0}_{-0.6}$ Vdc, +24.0 ±2.0 Vdc, and +30.0 $^{+0}_{-1.0}$ Vdc. Measure the output voltage at J1-7 for each input voltage condition, to determine compliance with 3.3.1.3.

4.6.5 Continuity test. With neither of the power sources (item 2 of table I) connected, perform resistance measurements between J1-5 and each of J1-2, J1-4, and J1-8, to determine compliance with 3.3.1.4.

4.6.6 Environmental tests

4.6.6.1 Temperature. The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that the assembly may be mounted on the inside surface of the test chamber.

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Perform this test in accordance with Procedure I of MIL-F-13926, with conditions C (-40°F) and E (+125°F), and with the rate of temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the tests of 4.6.2 through 4.6.5 to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.6.2 Vibration. Perform this test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Part 1 of procedure I therein except that the vibratory conditions of 3.3.2.3 shall apply. Following vibration exposure, examine the assembly for structural failures, loose hardware, wire breakage, and abrasions. Perform the tests of 4.6.2 through 4.6.5 to determine compliance with 3.3.2.3.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11737807. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The assembly is a module of the Laser Receiver-Transmitter RT-1021 ( )/VVG-1, intended for use in the Fire Control Rangefinder AN/VVG-1 ( ). The assembly functions to:

a. Provide a regulated +10 Vdc output voltage

b. Provide a regulated +15 Vdc output voltage.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

b. Applicable packaging data sheet number (see 5.1)

c. Selection of applicable levels of preservation, packaging, and packing

d. Applicable stock number

e. Requirement for First Article Submission.

CUSTODIAN:

Army - MU

PREPARING ACTIVITY

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PROJECT NO. 1240-A313

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-E-50745(MU) Electronic Component Assembly: 11737807 (Pre-Regulator)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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